

LEARNING INFRASTRUCTURE FOR THE NEXT BILLION CHILDREN

# NEXT BILLION CHILDREN

Programmable physical learning environments where children build,  
experiment and develop capabilities with shared AI.

NBC · built by Algo Peers · Sam Quansah, founder · Pre-seed · September 2026

## PROBLEM

# ANSWERS ARE GETTING CHEAPER. CAPABILITY ISN'T.

AI makes explanations and finished work easy to get. The chance to investigate, build, get it wrong and try again still costs time, materials and skilled adults.

### No equipment

A child in Cape Coast has few chances to experiment with real things.

### AI everywhere

A child with plenty of AI can't explain or repeat the result when the help changes.

### Unseen skill

A child shows real understanding through action, and no test records it.

## SOLUTION

# PROGRAMMABLE PHYSICAL LEARNING ENVIRONMENTS

### Programmable materials

micro:bit, sensors and everyday materials whose roles and rules change, so one kit supports many worlds.

### Learning Worlds

Designed learning experiences around real problems in the child's own world.

### Shared AI

One AI shared by a centre helps facilitators prepare, translate and give feedback. No device per child.

### Evidence of capability

What a child can explain, create and do when the task changes. Families can see and correct it.

**Learning comes first.** The experience has to be worth it for the child. Evidence supports the next learning decision, and a child's record is never the product.

WHAT A LEARNING WORLD IS

# A DESIGNED EXPERIENCE, NOT A WORKSHEET

## 1 · Belong

A story and a real problem from the child's world. "Do I see my world here?"

## 2 · Investigate

Children build with programmable materials and ask AI good questions.

## 3 · Understand

Predict, test, explain, revise. "What is really going on underneath?"

## 4 · Apply

Build something for someone real, then show the skill in a new situation.

Each world ships with a **facilitator guide, materials list, AI prompts and short checks**. Printed cards and tokens are only the low-cost floor for rooms with nothing else.

Phases follow Algo Peers' Root Access pedagogy.

EXAMPLE WORLD · ALREADY RUN BY ALGO PEERS

# FARM WORLD: TECHNOLOGY FOR FOOD SECURITY

## What children did

Built AI pest detection, smart irrigation, plant growth and soil moisture monitors, and a vegetable classifier, using micro:bit, Teachable Machine, recycled materials and cocopeat. Delivered in person, in online sessions and through video.

**51**

learners aged 7–15

**34**

communities

**53%**

girls

**7**

working prototypes

Algo Peers module, documented at [labs.algopeers.com](https://labs.algopeers.com). Participation and outputs, not a measured learning effect. NBC turns worlds like this into repeatable packages other facilitators can run.

## WHY NOW

# THREE THINGS CHANGED

### **AI access is spreading, unevenly**

Some children have more AI than they can use well. Many have almost none. Both need real practice.

### **Computing became compulsory**

Ghana's schools must now teach computing, and most have few working computers.

### **Shared AI is cheap enough**

One phone or computer can now serve a whole learning centre, offline or online.

## TRACTION

# FIVE YEARS OF PRACTICE. NBC STARTS FROM IT.

### **Algo Peers, since 2021 · Cape Coast, Ghana**

- 1,000+ children reached through our programmes
- Families pay for our after-school programmes
- \$93,511 in grants, including Global Affairs Canada
- 31 public schools assessed (11,214 students)
- Our own learning platform and educator training

### **NBC, started 2026**

- Thesis, three Learning Worlds and a draft task format
- Website and classroom prototypes
- No NBC pilots or revenue yet
- Algo Peers' results don't prove NBC works. The pilot will test that

FIRST CUSTOMER

# PROGRAMMES THAT ALREADY RUN HANDS-ON LEARNING

Independent operators of recurring hands-on programmes for children aged about 9–12, starting with reachable operators in Cape Coast.

| ROLE    | WHO                                                   |
|---------|-------------------------------------------------------|
| Uses it | Children in the programme                             |
| Runs it | A facilitator                                         |
| Buys it | The programme owner. Parents may fund it through fees |

**Offer:** NBC Learning Worlds. Reusable physical experiences, facilitator guidance and a short evidence summary that helps choose each child's next step.

## MARKET

# TAM, SAM, SOM

**\$29–53B**

### TAM

Global after-school programmes, 2025 (industry estimates; methods vary). 1.4 billion students are enrolled worldwide.

**\$99M**

### SAM

Ghana's ~27,400 primary and junior high schools × \$3,600 per site a year. Our first market.

**\$3.2M**

### SOM, year 5

800 paying sites (about 3% of Ghana's schools) in our central scenario.

Site price = 240 participants × \$15 (the thesis's own illustration). Schools: Ghana education sector data (2020). Enrolment: UNESCO (2024). Market estimates: several research firms (2025).

# ONE CORE BUSINESS, TWO OPTIONS

## Core

### Learning service

A programme or site fee from operators. A family subscription through our platform is a separate test.

## Option

### Shared AI service

Useful AI work for operators and communities, only if they pay for it again and again.

## Option

### Creators and licensing

Licensed experience packages and set-up services, and later, verification.

**The learning business must stand alone.** Our downside case assumes zero revenue from the options. We never sell children's data.

UNIT ECONOMICS

# HOW A SITE BREAKS EVEN

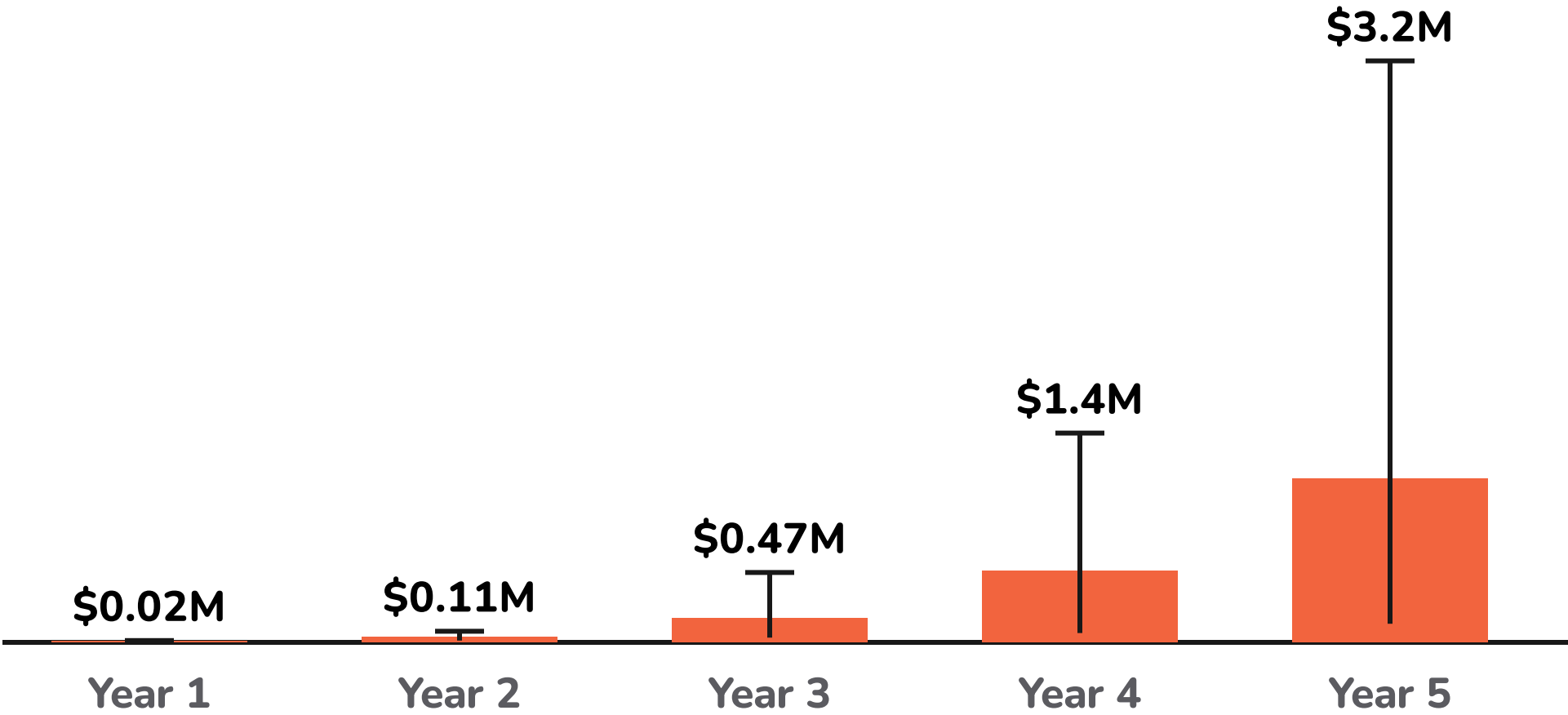
| PARTICIPANTS A YEAR | COST PER PARTICIPANT |
|---------------------|----------------------|
| 60                  | \$26                 |
| 120                 | \$16                 |
| 240                 | \$11                 |

Example: \$1,200 fixed delivery cost a year plus \$6 per participant. At a price of \$15, a site covers these costs at **134 participants**.

Illustration only, not our price or forecast. Real prices come from the pilot's cost sheet. Next: scenarios built on these numbers.

PROJECTIONS

THREE SCENARIOS, FIVE YEARS



| YEAR 5  | SITES | REVENUE |
|---------|-------|---------|
| Low     | 100   | \$0.36M |
| Central | 800   | \$3.2M  |
| High    | 2,500 | \$11.3M |

Bars: central scenario; lines: low to high. Revenue = sites × \$3,600, plus the options (0% low, 10% central, 25% high, from year 3). High adds a second country from year 3. Projections from stated assumptions, not forecasts. We rebuild them from pilot data.

## THE FUTURE

# FROM ONE WORLD IN CAPE COAST TO INFRASTRUCTURE

### 2026–27

Prove one world works without us, in Cape Coast and one contrasting site.

### 2028–29

A library of worlds, sponsored public schools, and a second country.

### 2030+

Shared AI for whole communities, and capability records children carry across schools and countries.

**Forecasts we track:** buyers will value the experience before the records · AI costs will fall, so we stay provider-neutral · every new site will need local adaptation · the learning business must stand on its own.

COMPETITION

# WHAT CHILDREN USE TODAY, AND WHAT WE MUST BEAT

|                                  | REAL HANDS-ON PRACTICE | WORKS WITHOUT A DEVICE PER CHILD | SHOWS WHAT A CHILD CAN DO |
|----------------------------------|------------------------|----------------------------------|---------------------------|
| Tablet and AI tutor apps         | No                     | No                               | Screen only               |
| Robotics and coding kits         | Yes                    | Costly                           | No                        |
| Exams and tests                  | No                     | Yes                              | Recall only               |
| Good materials + a great teacher | Yes                    | Yes                              | Informal                  |
| NBC                              | Yes                    | Yes                              | Yes                       |

Our real competitor is the last row. NBC has to make great hands-on learning cheaper to repeat and easier to run.

## MILESTONES

# EIGHT WEEKS TO A CLEAR DECISION

### Weeks 1–3

12–16 operator conversations. One capability, three task designs, a costed pilot offer. Offers to 10 qualified buyers.

### Weeks 4–6

External facilitators deliver it. A delayed, unfamiliar task, scored blind. Payments and full costs recorded.

### Weeks 7–8

A contrasting site if access is ready. Review every gate: repeat, simplify, change payer or pause.

**Pass lines, set before we start:** 3 of 10 buyers pay · 2 outside facilitators deliver within support · results are readable · scorers agree · each site covers its direct costs.

## RISKS

# WHAT COULD MAKE US WRONG

- **Substitution:** good materials and a great facilitator may do as well, more cheaply. Then we simplify to the part that helps.
- **Transfer** may not appear, or may cost too much to measure well.
- **Operators** may value the programme but not pay enough to cover delivery.
- **Trust:** automatic skill scoring could lose trust. Records stay human-readable, and NBC works without them.

## TEAM

# BUILT BY THE PEOPLE ALREADY IN THE ROOM

### Sam Quansah

#### Founder

Founded Algo Peers in 2021. Ed.M. candidate, Harvard GSE. 2024 Mandela Washington Fellow. Former Google Africa Developer Scholarship mentor.

### Vera Aninakwah

#### Learning Designer

Culturally responsive AI, computing and robotics for K-12. BBC micro:bit Champion and Africa Community Lead. Penn GSE scholarship recipient.

### Nana Adwoa Nsiah

#### Learning Experience Designer

Leads our Learning Media Lab: animation and games for African children. micro:bit Champion. NYU Steinhardt scholarship recipient.

**Hiring first:** a learning scientist, and a technical lead for physical computing and offline software.

# HOW CHILDREN WHO CAN'T PAY GET ACCESS

## Sponsored schools

Companies and foundations fund Learning Worlds in public schools. Our non-profit delivers them.

## Education authorities

License Learning Worlds and teacher training as computing becomes compulsory. After the pilot.

## Community centres

Libraries and makerspaces host kits and a shared AI. Local government gives space; sponsors fund kit.

## Universities

Co-design and independently evaluate our learning studies.

**Two entities:** Algo Peers Ltd builds and sells NBC. Algo Peers Impacts, our non-profit, delivers access. Sponsor money is ring-fenced, and children's data is never part of any deal.

## THE ASK

# RAISING A PRE-SEED ROUND

### The pilot

The eight-week programme with external facilitators, delayed tests and paid buyers.

### A second site

A contrasting setting, with different access to AI, for the same age group.

### The team

A learning scientist and a technical lead, plus a properly powered follow-up study.

Round details on request.